

Lab Tasks

Name: Muhammad Faisal

Roll no: 2022F-BCS-152

Section: A

Instructor Name: Miss Fozia Khan

Course: Artificial Intelligence Lab (CS - 325L)

Program: (BS) - Computer Science

Lab #08

Activity- 1:

Convert following sentences in to PROLOG FACTS and specify whether they represent RELATIONSHIP or PROPERTY:

Declarative Sentence	PROLOG FACT	RELATIONSHIP or PROPERTY
SSUET is university	university(ssuet).	% Property
Faheem likes ice-cream	likes(faheem, icecream).	% Relationship
Birds can fly	can_fly(bird).	% Property
Switch is off	off(switch).	% Property
Sara is nephew of Saima	nephew(sara, saima).	% Relationship

lab08 - Notepad

File Edit Format View Help

% Activity 1 Facts

```
university(ssuet).      % Property
likes(faheem, icecream). % Relationship
can_fly(bird).          % Property
off(switch).            % Property
nephew(sara, saima).    % Relationship
```

GNU Prolog console

File Edit Terminal Prolog Help

GNU Prolog 1.5.0 (64 bits)

Compiled Jul 8 2021, 12:33:56 with cl

Copyright (C) 1999-2021 Daniel Diaz

```
| ?- change_directory('D:/Sir Syed University/6th Semester/AI Lab').
```

yes

```
| ?- [lab08].
```

compiling D:/Sir Syed University/6th Semester/AI Lab/lab08.pl for byte code...

D:/Sir Syed University/6th Semester/AI Lab/lab08.pl compiled, 8 lines read - 832 bytes written, 6 ms

yes

```
| ?- university(ssuet).
```

yes

```
| ?- likes(faheem, icecream).
```

yes

```
| ?- can_fly(bird).
```

yes

```
| ?- off(switch).
```

yes

```
| ?- nephew(sara, saima).
```

yes

```
| ?- nephew(sara, x).
```

no

```
| ?- off(x).
```

no

```
| ?- nephew(sara, saima).
```

yes

```
| ?- nephew(sara, X).
```

X = saima

yes

Activity- 2:

Make prolog facts from following sentences and answer the questions by typing queries on prolog.

- Saleem is father of Aslam.
- Dania is mother of Aslam.
- Aslam and Saeed are brothers.
- Faiza is sister of Ali.
- Sana is enemy of Sumra.

```
% Activity 2 Facts
father(saleem, aslam).
mother(dania, aslam).
brother(aslam, saeed).
brother(saeed, aslam).
sister(faiza, ali).
enemy(sana, sumra).
```

Questions**1. List brothers.**

```
yes
| ?- brother(X, Y).

X = aslam
Y = saeed ?

(15 ms) yes
```

2. Who is father of Aslam?

```
| ?- father(X, aslam).

X = saleem

yes
```

3. Dania is mother of whom?

```
| ?- mother(dania, X).

X = aslam

yes
```

4. Who is brother of Aslam?

```
| ?- brother(X, aslam).

X = saeed

yes
```

5. Faiza is sister of whom?

```
| ?- sister(faiza, X).

X = ali

yes
```

Activity- 3:

Create a file and write below fact and execute this using below question.

```
has(jack, apples).
has(ann, plums).
has(dan, money).
fruit(apples).
fruit(plums).
```

Questions**1. List fruits.**

```
| ?- fruit(X).
X = apple ?
yes
```

2. Which fruit does Jack have?

```
| ?- has(jack, X).
X = apple ?
yes
```

3. Does Dan have something?

```
| ?- has(dan, X).
X = mango
yes
```

4. Who has apples and who has plums?

```
| ?- has(X, apple).
X = jack ?
yes
| ?- has(X, plum).
X = jack ?
yes
```

5. Does someone have apples and plums?

```
| ?- has(X, apple), has(X, plum).
X = jack ?
yes
```

6. Does Dan have fruits?

```
| ?- has(dan, X).
X = mango
yes
```

Activity- 4:

Write down the PROLOG Rules for the following sentences:

Declarative Sentences	PROLOG RULES
Person1 is sibling of Person2 if Person3 is parent of Person1 and Person3 is parent of Person2.	sibling(Person1, Person2) :- parent(Person3, Person1), parent(Person3, Person2), Person1 \= Person2.
Person1 is child of Person2 if Person2 is parent of Person1.	child(Person1, Person2) :- parent(Person2, Person1).
It will rain today if clouds are heavy and air pressure is low.	rain_today :- heavy_clouds, low_air_pressure.

Activity- 5:

A person gets good grades if he studies well and is regular in class. Write program in prolog to check whether a given person gets good grade or not?

```
% Activity 5 - Good Grades
studies_well(faheem).
studies_well(sara).
studies_well(ali).

regular_in_class(faheem).
regular_in_class(ali).

good_grades(Person) :-
    studies_well(Person),
    regular_in_class(Person).

| ?- good_grades(faheem).
yes
| ?- good_grades(sarah).
no
| ?- good_grades(ali).
no
```

Activity- 6:

Here are some declarative sentences convert them into PROLOG facts and answer following questions.

- Kashif likes music.
- Kashif likes cricket.
- Saba likes chips.
- Saba likes mobile.
- Kashif hates cold drinks.
- Saba hates driving.
- Saba hates parties.

```
% Activity 6 Facts
likes(kashif, music).
likes(kashif, cricket).
likes(saba, chips).
likes(saba, mobile).

hates(kashif, cold_drinks).
hates(saba, driving).
hates(saba, parties).
```

Questions**1. Does Saba like chips?**

```
| ?- likes(saba, chips).
```

```
true ?
```

```
(16 ms) yes
```

2. Does Kashif hate cricket?

```
| ?- hates(kashif, cricket).
```

```
no
```

3. Does Kashif like music?

```
| ?- likes(kashif, music).
```

```
true ?
```

```
yes
```

4. What does Kashif like?

```
| ?- likes(kashif, What).
```

```
What = music ?
```

```
yes
```

5. What does Saba like?

```
| ?- likes(saba, What).
```

```
What = chips ?
```

```
yes
```

6. Who likes music?

```
| ?- likes(Who, music).
```

```
Who = kashif ?
```

```
yes
```

Activity- 7:

There is a cricket team of 5 players; capabilities of each of the player are given in following table.

Player Name	Batsman	Bowler	Fielder
Salman	✓		✓
Afridi	✓	✓	✓
Malik	✓	✓	✓
Shoaib		✓	✓
Tanveer		✓	

Write the facts from the above table and create the rule for all Rounder and run following goals in PROLOG.

1. Is Afridi all-rounder?**2. Is Shoaib all-rounder?****3. Who are all-rounders in the team?**

% Activity 7 - Cricket Team Capabilities

```
batsman(salman).
batsman(afridi).
batsman(malik).
batsman(shoaib).

bowler(salman).
bowler(afridi).
bowler(malik).
bowler(shoaib).
bowler(tanveer).

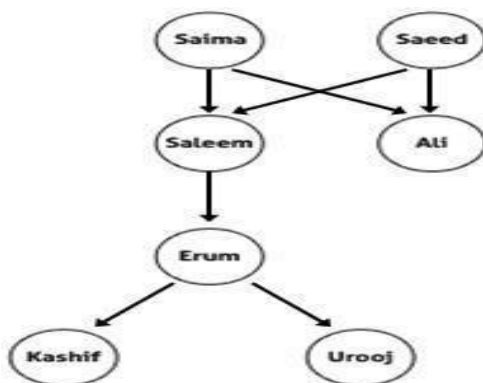
fielder(salman).
fielder(afridi).
fielder(shoaib).

% Rule: All-rounder is a player who is a batsman, bowler, and fielder
all_rounder(Player) :-
    batsman(Player),
    bowler(Player),
    fielder(Player).

| ?- all_rounder(afridi).
yes
| ?- all_rounder(shoaib).
yes
| ?- all_rounder(X).
X = salman ?
(16 ms) yes
```

Activity- 8:

Case Study: Family Tree: Consider the following family tree shown in figure below



% Activity 8 Facts

```

father(saeed, saleem). % Saeed is the father of Saleem
father(saeed, ali).    % Saeed is the father of Ali
father(saleem, erum).  % Saleem is the father of Erum

mother(saima, saleem). % Saima is the mother of Saleem
mother(saima, ali).    % Saima is the mother of Ali
mother(erum, kashif).  % Erum is the mother of Kashif
mother(erum, urooj).   % Erum is the mother of Urooj

```

% Rules

```

% Grandfather: X is grandfather of Y if X is father of Z and Z is parent of Y
grandfather(X, Y) :-

```

```

    father(X, Z),
    (father(Z, Y); mother(Z, Y)).

```

```

% Grandmother: X is grandmother of Y if X is mother of Z and Z is parent of Y
grandmother(X, Y) :-

```

```

    mother(X, Z),
    (father(Z, Y); mother(Z, Y)).

```

```

% Siblings: X and Y have same father and mother but are not the same person

```

```

brother(X, Y) :-
    father(F, X), father(F, Y),
    mother(M, X), mother(M, Y),
    X \= Y.

```

```

sister(X, Y) :-
    father(F, X), father(F, Y),
    mother(M, X), mother(M, Y),
    X \= Y.

```

```

% Direct parent relationships

```

```

father_of(X, Y) :- father(X, Y).
mother_of(X, Y) :- mother(X, Y).

```

Questions:

1. Is Saleem father of Ali?

```

| ?- father(saleem, ali).

```

```

no

```

2. Is Kashif brother of Urooj?

```

| ?- brother(kashif, urooj).

```

```

no

```

3. Who is father of Kashif?

```

| ?- father(X, kashif).

```

```

no

```

4. Is Erum mother of Urooj?

```

| ?- mother(erum, urooj).

```

```

yes

```

5. Who is grandfather of Urooj?

```
| ?- grandfather(X, urooj).
```

```
X = saleem
```

6. Is Salma grandmother of Erum?

```
| ?- grandmother(salma, erum).
```

```
no
```

7. Saeed is father of whom?

```
| ?- father(saeed, X).
```

```
X = saleem ? ;
```

```
X = ali
```

```
yes .
```

8. Salma is mother of whom?

```
| ?- mother(salma, X).
```

```
no
```

Activity- 9:

Create selection criteria using rules.

1. starter(soup).
2. starter(salad).
3. main_course(rice).
4. main_course(pizza).
5. main_course(burger).
6. desert(custard).
7. desert(ice_cream).

Create rules for hungry, very_hungry and on_diet.

- For hungry: only starter and main course
- For very_hungry starter, main course and desert
- For on_diet it displays only starters.

```
% Activity 9
```

```
% Starters
```

```
starter(soup).
```

```
starter(salad).
```

```
% Main Course
```

```
main_course(rice).
```

```
main_course(pizza).
```

```
main_course(burger).
```

```
% Desert
```

```
desert(custard).
```

```
desert(ice_cream).
```

```
% Hungry: starter and main course
```

```
hungry(Item) :-
```

```
    starter(Item);
```

```
    main_course(Item).
```

```
% Very Hungry: starter, main course and desert
```

```
very_hungry(Item) :-
```

```
    starter(Item);
```

```
    main_course(Item);
```

```
    desert(Item).
```

```

% On Diet: only starter
on_diet(Item) :-
    starter(Item).
| ?- hungry(X).

X = soup ? ;

X = salad ? ;

X = rice ? ;

X = pizza ? ;

X = burger

yes
| ?- very_hungry(X).

X = soup ? ;

X = salad ? ;

X = rice ? ;

X = pizza ? ;

X = burger ? ;

X = custard ? ;

X = ice_cream

(47 ms) yes
| ?- on_diet(X).

X = soup ? ;

X = salad

yes ,

```

Activity- 10:

Write a program that doubles any number that is taken from user as input.

```

% Activity 10
double_number :-
    write('Enter a number: '),
    read(Number),
    Double is Number * 2,
    write('Double is: '),
    write(Double), nl.

| ?- double_number.
Enter a number: 7.
Double is: 14

yes ,

```

Activity- 11:

Calculate Arithmetic mean of any three numbers.

% Activity 11

```

arithmetic_mean :-
    write('Enter first number: '),
    read(A),
    write('Enter second number: '),
    read(B),
    write('Enter third number: '),
    read(C),
    Mean is (A + B + C) / 3,
    write('Arithmetic Mean is: '),
    write(Mean), nl.

```

```

| ?- arithmetic_mean.

```

```

Enter first number: 5.

```

```

Enter second number: 10.

```

```

Enter third number: 15.

```

```

Arithmetic Mean is: 10.0

```

Activity- 12:

Create a basic calculator prolog program for making basic calculations (add, subtract, multiply, divide, average). Use separate goal statements for making each calculation (with constant numeric values).

% Activity 12

% Addition

add :-

```

    write('Enter first number: '), read(A),
    write('Enter second number: '), read(B),
    Result is A + B,
    write('Addition: '), write(Result), nl.

```

% Subtraction

subtract :-

```

    write('Enter first number: '), read(A),
    write('Enter second number: '), read(B),
    Result is A - B,
    write('Subtraction: '), write(Result), nl.

```

% Multiplication

multiply :-

```

    write('Enter first number: '), read(A),
    write('Enter second number: '), read(B),
    Result is A * B,
    write('Multiplication: '), write(Result), nl.

```

% Division

divide :-

```

    write('Enter numerator: '), read(A),
    write('Enter denominator: '), read(B),
    ( B =\= 0
    -> Result is A / B,
        write('Division: '), write(Result), nl
    ; write('Error: Division by zero is not allowed.'), nl
    ).

```

% Average of three numbers

average :-

```

    write('Enter first number: '), read(A),
    write('Enter second number: '), read(B),
    write('Enter third number: '), read(C),

```

```

Avg is (A + B + C) / 3,
write('Average: '), write(Avg), nl.

```

```

| ?- add.
Enter first number: 2.
Enter second number: 3.
Addition: 5

```

```

(16 ms) yes
| ?- subtract.
Enter first number: 3.
Enter second number: 1.
Subtraction: 2

```

```

yes
| ?- multiply.
Enter first number: 3.
Enter second number: 3.
Multiplication: 9

```

```

(15 ms) yes
| ?- divide.
Enter numerator: 2.
Enter denominator: 6.
Division: 0.33333333333333331

```

```

(15 ms) yes
| ?- average.
Enter first number: 2.
Enter second number: 3.
Enter third number: 4.
Average: 3.0

```

```

yes

```

Activity- 13:

Make a calculator program by using write() and read() predicates, for taking numeric input from user and displaying text after program execution.

```

% Activity 13|
calculator :-
    write('Enter the first number: '), read(A),
    write('Enter the second number: '), read(B),

    Sum is A + B,
    Diff is A - B,
    Prod is A * B,

    ( B =\= 0
    -> Div is A / B
    ; Div = 'undefined (division by zero)'
    ),

    Avg is (A + B) / 2,

    nl, write('--- Calculation Results ---'), nl,
    write('Addition: '), write(Sum), nl,
    write('Subtraction: '), write(Diff), nl,
    write('Multiplication: '), write(Prod), nl,
    write('Division: '), write(Div), nl,
    write('Average: '), write(Avg), nl.

```

```
| ?- calculator.  
Enter the first number: 2.  
Enter the second number: 1.  
  
--- Calculation Results ---  
Addition: 3  
Subtraction: 1  
Multiplication: 2  
Division: 2.0  
Average: 1.5  
  
yes
```